LEARNER GUIDE

Reading real parabolas

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

This is the last lesson of the unit, and it brings the parabola into the real world. Throw a ball and its height over time traces a parabola; so does a jet of water, a hanging chain, or a graph of a shop's revenue against its price. What makes these graphs useful is the turning point. For an upside-down curve (a ∩) the turning point is the highest point — a maximum; for a U it is the lowest point — a minimum. And it always means something: the greatest height a ball reaches and when; the price that earns the most money; the amount that costs the least to make. You will read the turning point from real, labelled graphs and say what it means; decide whether each turning point is a maximum or a minimum; and use a quadratic to find the biggest growing bed you can enclose with a fixed length of fencing. No calculator today. The question you are exploring is: where do U-shaped curves show up, and what do their turning points mean?

What you are learning to notice

- That real situations — a thrown ball, revenue against price, area against a side — often make parabolas.
- That a downward curve (∩) has a maximum turning point, and an upward curve (U) has a minimum.
- That the turning point's two coordinates are an answer: the most (or least) of something, and when or where it happens.
- That for a fixed length of fence, the area you can enclose is a quadratic with a maximum — a square encloses the most.

Moving through it, one step at a time

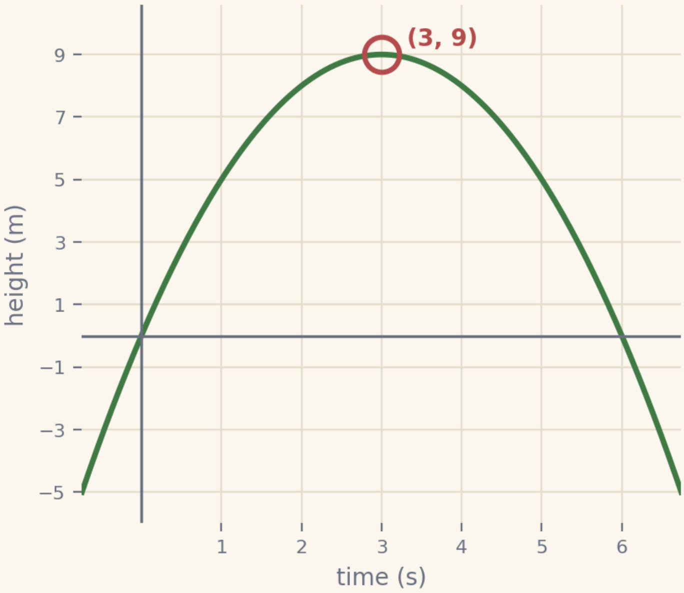
In The shape is everywhere, read the top of a thrown ball's arc. In Maximum or minimum?, decide which way each real curve opens and what that means. In The parabola in the world, read a turning point from labelled axes and interpret it. In The biggest growing space, move a slider to find the side length that gives the most area for 20 m of fencing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — the greatest height

A ball's height is $h = -(t - 3)^2 + 9$ (metres, against time in seconds). Find its greatest height and when it happens.

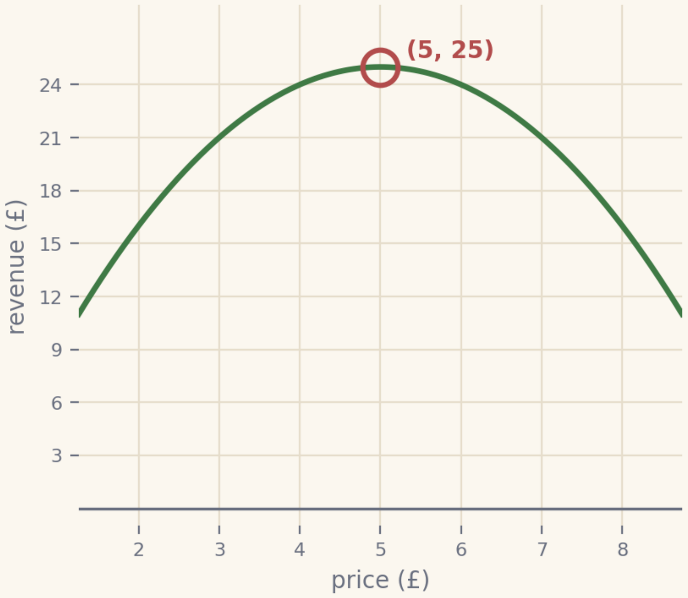


1. the curve opens	downward (∩), so the turning point is a maximum
2. the top is where the bracket is 0	$t - 3 = 0$, so $t = 3$
3. the height there	$h = 0 + 9 = 9$
4. so, in words	greatest height 9 m, reached at 3 s

the ball reaches a maximum height of 9 m at 3 seconds

Example 2 — the best price for the most revenue

A shop's revenue is R = –(price – 5)^2 + 25 (£, against price in £). Find the price that earns the most, and how much.



1. the curve opens	downward (∩), so the turning point is a maximum
2. the top is where the bracket is 0	price – 5 = 0, so price = 5
3. the revenue there	R = 0 + 25 = 25
4. so, in words	most revenue £25, at a price of £5

a price of £5 earns the most revenue, £25

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the parabola and its turning point (Lessons 5–7), reading coordinates from labelled axes, and substituting into a simple expression), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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